

The Boundary Is the Locus

Mathematics as a Contact Phenomenon, and the Sphere as Its Zero

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September, 2026

Abstract

The standard picture treats mathematics as a medium in which objects are situated and described. This paper inverts that picture. Mathematics does not fill space and does not reach into interiors. It occurs at boundaries, and only at boundaries, because a boundary is the only structure that supplies the distinctions any mathematical operation requires. The interior of a body is not poorly known — it is silent, in the precise sense that no coordinate, relation, or comparison is available there. The exterior is not a region with properties; it is the complement, defined only by reference to the boundary it lies outside of.

Under this inversion the sphere occupies a distinguished position that is not aesthetic and not conventional. It is the unique body specified by a single scalar with no orientation datum, that scalar being measured from the interior outward. It is the unique zero of the isoperimetric deficit $\delta(K) = A^3/(36\pi V^2) - 1$, which this paper identifies as the mathematical content of a shape. It has one boundary site where every polyhedron has many. And it is the unique convex body whose contact with any other convex body is generically zero-dimensional — the minimum possible aperture through which mathematics can enter.

Three classical results carry the argument and none of them is novel: the Jordan–Brouwer separation theorem, which forces exactly two regions from any closed surface without any choice being made; the isoperimetric inequality, which makes the sphere the unique minimiser of boundary per volume; and the transitivity of $SO(3)$ on the sphere, which removes every orientation datum. What is new is the reading. Each of these says the same thing once the inversion is applied: the sphere is the shape that admits the least mathematics, and departure from sphericity is exactly mathematical content.

The consequences follow. Matter is a stabilised transformation boundary and its mathematics is emergent from that boundary rather than resident in any substance. Transformations are selected, not created: a shape's admissible transformations are a property of its boundary and exist whether or not any is taken. Calculation is downstream of transformation and requires distinctions transformation does not, so transformations occur that no mathematics can express. Every claim is graded, and the closing sections state what must follow next rather than what this paper declines to say.

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1. The inversion

Mathematics is normally treated as ambient. Space is imagined as already coordinatised, objects are placed into it, and their properties are read off using machinery that was there before they arrived. Under that picture the interior of a body is as mathematically populated as anywhere else — it has coordinates, it has a metric, one can integrate over it — and the boundary is merely the place where one body's properties stop and another's begin.

This paper takes the opposite position. Mathematics is not ambient and does not precede the objects. It occurs where things touch, and nowhere else.

The claim is not that interiors are difficult to access. It is that an interior supplies nothing for a mathematical operation to act on. Every operation requires a distinction: a comparison needs two things, a derivative needs a neighbourhood with directional structure, a coordinate needs an origin, a measure needs a boundary to measure up to. Inside a homogeneous body none of these is available, and their absence is not a limitation of instruments. It is what interior means.

1.1 What the inversion forbids

Stating the prohibitions first prevents the argument from silently reverting. Under this inversion the following moves are not available and are not used anywhere in what follows:

- Placing a body into a pre-existing coordinate system and reading properties from that system. The coordinate system is a product of boundaries, not a container for them.
- Treating an interior point as a location. Interior points are not distinguishable from one another; naming one requires a boundary to name it relative to.
- Treating the exterior as a region with content. The exterior is unbounded and carries no parameter of its own.
- Deriving a shape from an equation. The equation is downstream: it is what becomes writable once the shape has supplied the distinctions the equation's symbols refer to.

The fourth prohibition is the strongest and the one most easily violated. It says that $x^2 + y^2 + z^2 = r^2$ is not what a sphere is. That equation presupposes three coordinate axes, an origin, and a metric — none of which a sphere supplies and all of which must be imported before the equation can be written. The equation describes a sphere that has already been placed inside a mathematical structure the sphere did not provide. [DERIVED]

2. The compilation order

Before anything is said about the sphere, the order in which mathematics becomes available has to be fixed, because every later section depends on it and because the natural way of writing violates it.

TRANSFORMATION -> REFLECTION -> BOUNDARY -> MATHEMATICAL SPACE -> FORMULA -> VALUE

Read downward this is compilation. Read upward it is de-compilation, and de-compilation is the direction of this investigation: a value is given, and the work is to recover the formula it renders, the relation the formula requires, the boundary that supplies the relation, and the transformation that produced the boundary.

VALUE -> FORMULA -> RELATION -> BOUNDARY -> REFLECTION -> TRANSFORMATION

2.1 Nothing in the later sections exists at the origin

This must be stated flatly because the paper's own notation invites the opposite reading. The quantities used throughout — r , A , κ , δ , θ , ϕ , the symmetry group and its dimension — are none of them available before contact. They are products of mathematical space having grown, and mathematical space grows only where a boundary relation exists.

An unmarked sphere does not have a radius. It does not have an area. It does not have a curvature. It is not that these are unknown or unmeasured; there is no relation present for them to be quantities of. Writing $4\pi R^2$ requires a reader who has already acquired a length, a ratio, a squaring operation, and a constant — every one of which is downstream. [DERIVED]

The centre is the sharpest case and the one this paper originally got wrong. The centre is defined as the point equidistant from the boundary. That is a statement about the boundary, so the centre is a boundary construction, and the radius measured from it is a boundary quantity. An earlier draft asserted that the centre requires no choosing and is therefore intrinsically available. It is not. It requires the boundary, and the boundary is what everything else is waiting on. [FALSIFIED — corrected here]

2.2 What the later quantities actually report

With the ordering fixed, the sections that compute r , δ , and the aperture count are not describing what a sphere contains. They are describing what a reader returns when it is applied to a sphere — and the answers are small precisely because the sphere supplies so little for a reader to work with.

So $\delta(\text{sphere}) = 0$ is not the claim that a sphere possesses zero mathematical content as an intrinsic attribute. It is the claim that a reader equipped to compute δ , applied to a sphere, finds nothing. The conclusion is unchanged and its status is corrected: these are reader results, not intrinsic properties, and the paper reads better for saying so. [DERIVED]

2.3 The math-free condition

The state prior to any boundary relation is not a state with few mathematical properties. It is a state with none, and the word for it here is math-free.

Math-free is stronger than 'no mathematics found.' It says mathematics has no coordinate structure through which to operate — not that the operations fail, but that they have no arguments. This is the same distinction the framework draws elsewhere between zero and absence: zero is a produced value and requires the machinery that produces it; math-free is the condition before any such machinery exists. [INTERPRETATION]

The sphere is the shape of that condition. It is not an object inside mathematical space with an unusually short property list. It is what the math-free condition looks like when a later reader, equipped with geometry, turns around and asks what shape was there.

2.4 The first event

Two undifferentiated spheres can do exactly one thing to each other: touch.

That contact is not yet a distance, a force, a curvature, a coordinate, or a topology. Each of those requires structure the contact has only just begun to supply. What the contact is, is the first relational distinction — each sphere being the other's stopping boundary, and neither possessing the relation alone.

math-free -> contact -> boundary relation -> first mathematics

Everything the remainder of this paper computes sits to the right of that arrow. [DERIVED]

2.5 The corpus-wide question this fixes

The ordering supplies a single question that can be asked of any result in this program, and asking it converts a collection of parallel observations into one de-compilation sequence:

What transformation creates the boundary that permits this mathematics to exist?

A result that cannot answer it has been stated at the wrong layer — it has imported a mathematical structure whose enabling boundary was never identified. That is the same error as describing a sphere by embedding it in a plane, and it recurs at every scale of the work. [DERIVED]

3. The sphere: what a reader returns

Among bodies in three dimensions the sphere holds a position that can be stated without reference to any coordinate system and without any aesthetic appeal.

3.1 The parameter count

A reader capable of specifying bodies up to rigid motion needs a certain amount of data for each. The count below is a property of that reader's task, not of the bodies in isolation — per Section 2, the bodies supply nothing until a boundary relation exists.

Body	Scalars	Orientation data	Total
Sphere	1 (r)	none	1
Cube	1 (edge)	3 angles	4
Cylinder	2 (r, h)	2 (axis direction)	4
Torus	2 (R, r)	2 (axis direction)	4
Ellipsoid	3 (semi-axes)	3 angles	6

The sphere is the unique body requiring one scalar and no orientation datum. This is forced rather than observed: $SO(3)$ acts transitively on directions, so no direction is distinguishable from any other, so no orientation datum can carry information. Any body that requires an orientation to be specified has already broken that transitivity — which is to say, it has a distinguished direction, which is a distinction, which is mathematics. [THEOREM]

3.2 The direction of the scalar

The single scalar is a radius, and a radius is measured from the centre outward. This is not a convention that could be reversed. There is no exterior point from which a distance to the sphere is defined without first choosing that point, and choosing it is an added distinction. The centre requires no choosing — it is the unique point equidistant from every boundary point, and its uniqueness is a consequence of the symmetry rather than a stipulation.

So once contact has supplied the relation, the sphere's only datum runs inward. That is the direction of the scalar, not evidence that it exists before the boundary does — the centre is defined by equidistance from the boundary and is therefore a boundary construction, per Section 2.1. What is asymmetric is that the inward measure terminates and the outward one does not. [DERIVED]

3.3 What is not available

Set against that single scalar, the following are unavailable on an unmarked sphere, and their unavailability is structural rather than practical:

- A distinguished point. Transitivity removes it.
- A distinguished direction. Same cause.
- An angular coordinate. θ and ϕ require a pole and a meridian, which are two marks that have not been made.
- A global flat chart. S^2 is not homeomorphic to the plane; any latitude-longitude system has a singularity where longitude ceases to be defined.
- An origin from which to begin counting anything.

The list is not a catalogue of the sphere's deficiencies. It is a description of what a body looks like when it has not yet touched anything. [DERIVED]

4. Duality is forced, not chosen

The first structure any closed surface produces is a division into two, and it produces it without any operation being performed and without any choice being made.

Jordan–Brouwer separation theorem. An embedded closed surface in three-dimensional space divides the complement into exactly two connected components, each having the surface as its full boundary. One component is bounded; the other is not.

Three features of this deserve to be stated separately because they are usually collapsed.

First, the number is exactly two. Not at least two, not generically two — two, always, for any closed embedded surface, however convoluted. The duality is not a feature of the sphere in particular but of closure in general. [THEOREM]

Second, nothing was added to obtain it. The surface was not cut, marked, coordinatised, or measured. Closure alone forces the bifurcation. This is the cleanest available instance of shape producing mathematics rather

than mathematics describing shape: the two-ness is a consequence of the shape's topology and could not have been derived from any equation written about the surface, since writing that equation already presupposes the ambient space the theorem is about. [DERIVED]

Third, and most consequential here, the two are not symmetric. Inside is bounded. Outside is not.

4.1 The asymmetry is not a sign flip

It is tempting to read inside and outside as a signed pair — the same thing with opposite orientation — and the temptation should be refused, because there is a genuine signed pair sitting nearby and conflating them costs the argument its precision.

The genuine signed pair is the surface normal. At any boundary point, n and $-n$ are the same relation read from the two sides, equal in magnitude and opposite in orientation. That is a phase pairing, and it is real.

Bounded versus unbounded is not that. It is not the case that the exterior is the negative of the interior. One region has finite extent and the other does not, and no reversal of sign converts between them. Magnitude asymmetry and phase asymmetry coexist on the same boundary and are different facts. [DERIVED]

5. The interior is silent

Take the bounded component. What is available inside it?

Every interior point of a homogeneous body stands in the same relation to the body as every other. There is no operation intrinsic to the interior that distinguishes one from another, because any such operation would require a reference, and every candidate reference is either the boundary — which is not interior — or another interior point, which is not distinguishable to begin with.

The centre appears to be an exception and is not. The centre is defined as the point equidistant from the boundary. That definition is a statement about the boundary. Remove the boundary and the centre is not merely unknown; it is undefined. Every interior structure that seems to be available on inspection turns out to be a boundary structure in disguise. [DERIVED]

5.1 Silent is not empty

This must be held carefully because the natural reading is the wrong one. The interior is not void, not vacant, not lacking substance. There is no shortage of being inside a body. What is absent is not material but distinction — the interior supplies no relation, and mathematics is made of relations.

The distinction between empty and silent is the same distinction the framework elsewhere insists on between zero and the empty set. Zero is a produced value. Silence is the absence of anything to produce a value from. A body's interior is maximally present and minimally sayable. [INTERPRETATION]

6. The exterior is nothing

Take the unbounded component. What is available there?

Nothing at all, and in a stronger sense than for the interior. The interior at least has one datum defined in terms of it — the radius, measured outward from the centre to the boundary. The exterior has no parameter

of its own whatsoever. It is not a region with extent, because it is unbounded and has no scale. It is not a region with a shape, because it is the complement of one. Everything sayable about the exterior is a statement about the boundary, phrased negatively.

This is where the inversion pays its largest dividend and also where it is most easily misread. The claim is not that the exterior is empty of matter. It is that the exterior is empty of mathematics: the outside of a body is where that body's mathematics has stopped, and it has no mathematics of its own to replace it with. [DERIVED]

6.1 Why the outside cannot be measured from

An observer standing outside a body and reporting on it is always reporting a boundary quantity. Distance to the surface, apparent size, direction — every one of these is a relation between the observer's own boundary and the body's, and the space between contributes nothing except its own emptiness of structure.

There is no exterior vantage that returns a value. Values come from inside, from the bounded region where a scale exists. This is why the radius is the sphere's only datum and why it is measured inward: the bounded side is the only side with a floor. [DERIVED]

7. Mathematics is a contact phenomenon

Assemble the previous three sections. The interior is silent, the exterior is nothing, and the boundary is what divides them. Everything sayable is therefore said at the boundary — not because boundaries are especially interesting, but because there is nowhere else.

Every formula in geometry is a boundary formula. Area is boundary. Curvature is boundary — it is the second-order failure of the boundary to be flat. Volume appears to be an interior quantity and is not: it is defined as what the boundary encloses, and the divergence theorem states this precisely, converting any volume integral into a boundary integral whenever the integrand permits. Distance is the relation between two boundaries. There is no formula in the subject that is genuinely about an interior. [DERIVED]

7.1 Contact rather than description

Two bodies in contact meet at a boundary that belongs to neither and to both. At that locus each is the other's boundary, and the mathematics available there is not a property of either body separately. Neither has it alone.

This is why contact is not measurement. Measurement suggests an instrument reaching in and extracting a value that was already present. At a contact, nothing is extracted, because there was nothing inside to extract. The relation comes into existence at the touch and did not exist as a hidden quantity beforehand. [INTERPRETATION]

7.2 The consequence for scale

If mathematics accumulates at boundaries and not in interiors, then the mathematical content of a body does not scale with its size.

An isolated sphere has one datum whether its radius is a micron or a light-year. Nothing is gained by growing it. What increases the mathematics available is not extent but contact — more boundary, more interfaces, more places where something stops against something else.

This inverts the usual intuition, which associates significance with magnitude. Under the present reading a body is not more mathematically substantial for being large. It is more substantial for being touched. [DERIVED]

8. The measure: mathematical content as isoperimetric deficit

If mathematics accumulates at boundaries, and if the sphere is the shape with the least boundary for its extent, then the sphere admits the least mathematics — and the quantity measuring how much a shape departs from the sphere measures how much mathematics it admits.

That quantity already exists and is standard.

Isoperimetric inequality. For any body K in three dimensions with surface area A and volume V , $A^3 \geq 36\pi V^2$, with equality if and only if K is a sphere.

Normalise it to a deficit:

$$\delta(K) = A(K)^3 / (36 \pi V(K)^2) - 1 \quad \geq 0, \quad = 0 \text{ iff sphere}$$

This paper reads δ as the mathematical content of a shape. The reading is new; the quantity is not. [DERIVED]

8.1 The ordering, computed

Computed exactly from closed-form area and volume:

Body	Area	Volume	δ
Sphere	12.5664	4.1888	0.000000
Regular icosahedron	8.6603	2.1817	0.206567
Regular dodecahedron	20.6457	7.6631	0.325034
Cylinder, $h = 2r$	18.8496	6.2832	0.500000
Regular octahedron	3.4641	0.4714	0.653987
Cube	6.0000	1.0000	0.909859
Cone, $h = 2r$	10.1664	2.0944	1.118034
Regular tetrahedron	1.7321	0.1179	2.307973
Torus, $R = 3r$	118.4353	59.2176	3.188790
Cylinder, $h = 20r$	131.9469	62.8319	4.145000

The ordering is not arbitrary and it is not a ranking of complexity in any informal sense. It tracks boundary per unit of being. The Platonic solids appear in descending order of face count — icosahedron before dodecahedron before octahedron before cube before tetrahedron — because fewer faces means each face

must be larger and flatter, and flatness is departure from the sphere. Elongation costs more than faceting: a cylinder ten times longer than it is wide carries more deficit than a tetrahedron. [COMPUTED]

8.2 What the zero means

$\delta = 0$ is attained by the sphere and by nothing else. Under the reading proposed here that is the statement that the sphere is the unique body admitting no mathematics beyond its single scalar.

The conventional reading treats mathematics as an ambient language applied to geometry. That relation is reversed here: geometry determines which mathematical relations become available, and the isoperimetric inequality has been reporting the extremal case of that fact since Steiner. The minimiser of boundary is the minimiser of mathematics, because boundary is where mathematics is. [DERIVED]

9. The discrete measure: aperture sites

For polyhedra there is a second, integer-valued version of the same quantity, and it separates the sphere from every polyhedron more sharply than δ does.

Body	Vertices	Edges	Faces	Aperture sites
Sphere	0	0	1	1
Tetrahedron	4	6	4	14
Cube	8	12	6	26
Octahedron	6	12	8	26
Dodecahedron	20	30	12	62
Icosahedron	12	30	20	62

The sphere has one face, no edges, and no vertices. It is the only body in the table with a single aperture site.

Euler's formula $V - E + F = 2$ holds for every entry, sphere included. The topological invariant is identical across the whole table. What differs is the site count, and the site count is not a topological invariant — it is a property of how the boundary has been divided. Two bodies with the same topology can admit vastly different amounts of mathematics because one has been faceted and the other has not. [COMPUTED]

9.1 A face is an aperture, an edge is a sharper one

A flat face admits planar geometry: coordinates on the face, distances within it, angles between features on it. None of that is available on a curved surface without additional structure. An edge admits more, because it is a locus where two planar systems meet and their disagreement becomes measurable. A vertex admits more still — three or more systems meeting at a point.

This is why the site count rises faster than δ for heavily faceted bodies, and why the two measures order the Platonic solids differently. δ measures how much extra boundary exists; the site count measures how many distinct places mathematics can enter. They are related but not the same, and neither reduces to the other. [DERIVED]

10. Contact: the minimum nonzero aperture

If an isolated sphere admits the least mathematics, the natural next question is what the least possible increase is — the smallest aperture that a contact can open.

Compute the dimension of the contact set for pairs of convex bodies:

Contact	Dimension of contact set
Sphere against sphere	0 — a point
Sphere against plane	0 — a point
Cylinder against plane	1 — a line
Cube against cube, corner to face	0 — a point
Cube against cube, edge to face	1 — a line
Cube against cube, face to face	2 — a square

The sphere is the only body whose contact with any convex body is generically zero-dimensional. A cube can meet another cube at a point, but only in the non-generic corner configuration; perturb it and the contact becomes a line or a face. A sphere cannot be perturbed into a higher-dimensional contact, because it has no flat region and no straight edge to lie along. [DERIVED]

10.1 The two bounds meet

The same property produces both results. The sphere has no flat region, which is why its boundary is minimal for its volume ($\delta = 0$), and it is also why its contact with anything is a single point (minimum aperture).

So the body that most completely refuses mathematics is also the body whose first contact admits the least mathematics that any contact can admit. That is not two coincident facts. It is one property read at two scales — the shape's own boundary, and the boundary it forms with another. [DERIVED]

11. The cost of a mark

Once contact has occurred, the accumulation of mathematics can be accounted for exactly. Each mark on the sphere consumes symmetry, and the symmetry is finite, so the ledger closes.

Marks	Continuous symmetry remaining	What has become available
0	SO(3), dimension 3	r only
1	SO(2), dimension 1	an axis; the equator is still one orbit
2	trivial, dimension 0	θ and ϕ ; a great-circle plane
3	trivial, mirror removed	chirality; the frame is fixed
4+	already trivial	content within an established frame, no new frame

Three facts in this table are forced rather than measured. The first mark leaves a full circle of rotational freedom — it establishes an axis, not an angle, so all directions around it remain in a single orbit and remain mutually indistinguishable. The second mark does not select an angle; it creates the angle, since before it there is no angle to select. And the third mark resolves a discrete ambiguity rather than a continuous one: two marks leave a reflection through the plane containing them and the centre, and only a third mark off that plane removes it. [THEOREM]

The ledger saturates at three. $\dim SO(3) = 3$, spent two on the first mark, one on the second, and one discrete bit on the third. A fourth mark adds no frame — it adds content within a frame that already exists. This is the sense in which a body's capacity to carry mathematics is finite and exactly countable: it is the dimension of the symmetry group its shape possesses, and marks spend it until it is gone. [DERIVED]

11.1 Antipodal marks cost nothing

Two marks at opposite poles leave the stabiliser at dimension one, not zero. They lie on the same axis, so the second supplies no information the first did not.

What is spent is not the count of marks but the amount of symmetry each one consumes. A mark placed on structure that already exists is free and produces nothing. This is the geometric statement of a principle that recurs throughout: a distinction that does not distinguish is not a distinction. [DERIVED]

12. Transformation precedes calculation

Everything to this point concerns where mathematics can occur. This section concerns what it is downstream of.

A calculation is a read operation. It requires a domain in which distinctions can be held apart long enough to be compared: something like space, so that two things can be elsewhere from each other; something like duration, so that a before and an after can be set against one another; and something capable of holding the comparison. Remove any of these and the calculation cannot be performed — not slowly, not approximately, at all.

A transformation requires none of them. $S \rightarrow S'$ needs no coordinate system to occur in, no clock to occur during, and no reader to register it. It is not a statement about a state; it is the state ceasing to be that state.

TRANSFORMATION $\rightarrow$ REFLECTION $\rightarrow$ DISTINCTION $\rightarrow$ QUERY SPACE $\rightarrow$ CALCULATION

Calculation sits at the far end of that chain, and each arrow adds a requirement. This is the ordering the rest of the paper has been assuming, stated explicitly. [DERIVED]

12.1 A transformation can exist without being representable

The consequence is immediate and it is the load-bearing one for the whole framework.

If calculation requires distinctions that transformation does not, then transformations occur that no available mathematics can express. Not hidden transformations, not transformations too small or too fast to detect — transformations for which the relational structure a description would need does not exist.

This is what an unmarked sphere is. It is not a mathematical object with a short list of properties. It is a real transformation with no relation available to state anything about it. It does not calculate its radius. It does not calculate its area. It transforms, and nothing is said. [DERIVED]

12.2 Mathematics grows toward what is already happening

Reality does not wait for mathematics in order to transform. Mathematics grows until it can read transformations that were already occurring.

That inverts the usual direction of explanation, in which mathematics is prior and reality conforms to it. Here mathematics is the trailing edge: the accumulated set of relations that boundaries have made available, always behind the transformation it describes, and growing only where a new boundary has opened. [INTERPRETATION]

It also dissolves the question of why mathematics applies to the world. It is not tracking the world from outside. It is the same event as the world touching itself, read at the place where the touching happens. There is no correspondence to explain because there are not two things to correspond. [INTERPRETATION]

13. Shape selects the mathematics

If mathematics occurs at boundaries, then a shape does not merely have mathematical properties. It determines which mathematics is reachable through it.

Written as a selection:

$$M(S) = \{ \text{operations whose required distinctions } S \text{ supplies} \}$$

A sphere supplies one scalar and no orientation, so $M(\text{sphere})$ is nearly empty. A closed curve supplies position modulo its length, so M admits modular arithmetic and closure but no depth. A nested bracket stream supplies position and depth as independent coordinates, so M admits matching, trees, and everything counted by the Catalan numbers — none of which is reachable on a flat sequence of the same symbols.

Shape	Coordinates supplied	Mathematics reachable
Sphere, unmarked	one interior scalar	the scalar; nothing else
Flat sequence	position	ordering, adjacency, difference
Closed curve	position mod n	modular arithmetic, closure
Nested stream	position and depth	matching, trees, Catalan structure
Faceted solid	faces, edges, vertices	planar geometry per face, angle at each edge

Nothing has been added to the substrate in moving down that table. The same symbols, the same material, arranged into different shapes, and each shape opens a different set of admissible operations. The shape is the access key. [DERIVED]

13.1 The instance in a discrete substrate

The same selection appears in a purely discrete setting, where it can be computed rather than illustrated. Consider a linear recurrence whose terms are formed from earlier terms at a fixed set of lags L . The recurrence is one formula. The lag set is a shape imposed on it.

For $L = \{1, 2\}$, the reachable set is generated with no gaps: every distance is representable, path parity is both necessary and sufficient for cancellation, and no ordering effects ever arise. For $L = \{2, 7, 15, 16\}$, the reachable set has gaps at 1, 3, and 5 — the gaps of the numerical semigroup $\langle 2, 7 \rangle$, with Frobenius number 5 — parity becomes necessary but not sufficient, and ordering becomes load-bearing at a computable threshold.

Same formula. Different shape. Different accessible mathematics, and the difference is exact and derivable from the shape alone. [COMPUTED]

14. Matter is a stabilised transformation boundary

Bring the preceding sections together and the position on matter follows without an additional assumption.

A body's mathematics is at its boundary. Its interior is silent. Its exterior is nothing. What remains of the body, once interior and exterior are subtracted, is its boundary — which is to say, its shape. There is no residue of substance left over that the shape is a shape of. [INTERPRETATION]

The derivation is short. Nothing about a body's interior enters any mathematics, so no mathematical account of a body ever requires interior substance as a term. Nothing about its exterior enters either, since the exterior carries no parameter of its own. What remains is the boundary, and the boundary is what a transformation produced and holds. A body is its boundary, held stable. [DERIVED]

14.1 The wrench

Melt a wrench and the wrench is gone. Nothing was destroyed: the same material is present, at the same mass, in the same place. What ended was a shape, and with the shape, a set of admissible transformations — a wrench can turn a bolt, and the melt cannot.

Under the reading here that is not a loss of substance dressed as a loss of form. It is the whole of what happened. The wrench was the shape; the shape ended; the material persisted and its potential returned to a state from which other shapes remain reachable. Form is not conserved. Potential is. [INTERPRETATION]

15. The dual existence

Every event in this account exists twice, and the two existences are not a thing and its description.

	Transformation	Mathematics
Requires	nothing	a boundary
Occurs	unconditionally	at contact only
Location	everywhere, including interiors	boundaries only

	Transformation	Mathematics
Observer	not required	not required, but a relation is
Can fail to exist	no	yes — routinely

The right-hand column is not a report about the left. It is the left-hand column occurring at a place where it can be stated. The same event, in an interior, is transformation and nothing else. The same event, at a boundary, is transformation and mathematics simultaneously.

This is the dual existence: not matter and mind, not object and description, but one process which is unconditional in itself and conditional in its sayability. The condition is contact. [INTERPRETATION]

15.1 Why the pair is asymmetric

The two sides do not have equal standing and the asymmetry has already been established twice. Jordan–Brouwer gives bounded against unbounded, which is not a sign flip. The interior carries one datum and the exterior carries none, which is not a sign flip either.

So the dual is not a mirror pair. Transformation is prior, unconditional, and complete; mathematics is posterior, conditional, and partial. Reading them as symmetric halves would restore exactly the ambient-mathematics picture this paper is inverting. [DERIVED]

16. Transformations are selected, not created

A shape's admissible transformations are a property of its boundary. They are therefore present whether or not any of them is taken.

Wood admits being made into a table. That admissibility is a fact about the wood's structure — its grain, its rigidity, its capacity to be cut and joined — and it holds in a forest with no carpenter anywhere. The carpenter does not create the table. The carpenter selects a transformation that was already admissible and performs it.

The table is then a new shape, and a new shape admits different transformations. A table lifts things off the floor. Lifting changes what is reachable — objects at table height are available to a standing body in a way objects on the floor are not. That change opens further transformations, which open further ones. None of them was invented at any step. Each was admissible before it was taken and became reachable when the preceding shape changed. [DERIVED]

16.1 The chain is a shape sequence, not a value sequence

Nothing in that chain is carried forward as stored information. There is no ledger inside the table recording that it was once wood. What is carried forward is constraint: the table's shape determines what can be done to it next, and that shape is the residue of what was done to it before.

This is memory without a memory register. The past is present as the current set of admissible transformations, and that set is exactly what the current shape supplies. [DERIVED]

16.2 What this does to the question of origin

If transformations are selected rather than created, then asking what caused a given transformation to occur is a different question from asking how it became possible. The second has an answer in the shape. The first may not have an answer at all, and the framework does not require one.

The set is complete. Nothing is added to it. What changes is which part of it a given configuration can reach. [INTERPRETATION]

17. The query space must remain open

One constraint from the wider framework is restated here because this paper sharpens it.

The original form — that every distinction must retain the potential to change — was read as a requirement on objects: that each thing must keep moving. That reading is now unnecessary, because transformation was shown in Section 11 to be unconditional. Objects do not need to be required to change; change is not something they can decline.

What the constraint must mean instead is that the set of reachable relations does not close. If the reachable set ever became fixed — every possible relation already available, no new boundary able to open — then no new mathematics could appear and the system would be terminal in the only sense that matters. Transformation would continue and nothing further would ever become sayable about it.

$Q(n+1)$ strictly contains $Q(n)$ for the system to remain non-terminal

So the continuation requirement is a statement about the query space rather than about the objects in it. This is a correction to the earlier reading, not an elaboration of it. [DERIVED]

18. What must follow

Four things are forced by the argument and not carried out here. They are obligations, not disclaimers, and each has a definite shape.

First, the general theorem. That every geometric quantity of a body is rewritable as a boundary quantity is used throughout and proved here only case by case. The divergence theorem supplies it wherever the integrand admits, which is why it survives every case examined. A general proof is this paper's principal debt and it is a well-posed problem.

Second, higher dimensions. δ and the aperture count are computed here in three. The isoperimetric inequality holds in every dimension with the ball as unique extremal case, so the account extends; what is uncomputed is how the two measures diverge as dimension rises, and whether they ever order bodies identically above three.

Third, non-convex bodies. Contact dimension is computed here for convex pairs. A non-convex body can meet another at a disconnected set, raising aperture count without raising δ proportionally, so the two measures should come apart. That divergence is the sharpest available test of whether they measure the same thing.

Fourth, the dynamical bridge. If matter is a stabilised transformation boundary and mathematical content is boundary excess, then a change in a body's boundary is a change in what can be said about it and in what transformations remain admissible to it. Whether that accounting connects to physical energy is a separate

derivation with its own obligations. It is the next thing this framework owes, and it is stated as a target rather than a boundary of the subject. [OPEN]

19. Claim ledger

§	Claim	Grade
2	Compilation order: transformation precedes formula precedes value	DERIVED
2.1	No quantity in this paper exists at the origin; all are post-contact	DERIVED
2.1	The centre requires no choosing (earlier draft)	FALSIFIED
2.3	The math-free condition is stronger than absence of found mathematics	INTERPRETATION
2.4	Two undifferentiated spheres can only touch; that is the first relation	DERIVED
3.1	The sphere is the unique body needing one scalar and no orientation	THEOREM
3.2	That scalar runs inward, measured from the centre outward	DERIVED
4	A closed surface forces exactly two regions, one bounded	THEOREM
4.1	Bounded/unbounded is a magnitude asymmetry, not a sign flip	DERIVED
5	No relation is available in a homogeneous interior	DERIVED
5.1	Silent is not empty; the interior lacks distinction, not being	DERIVED
6	The exterior carries no parameter of its own	DERIVED
7	Every geometric formula is a boundary formula	DERIVED
7.2	Mathematical content does not scale with extent, only with contact	DERIVED
8	$A^3 \geq 36\pi V^2$ with equality iff sphere	THEOREM
8.1	δ ordering across ten bodies, computed exactly	COMPUTED
8.2	δ is what a reader equipped to compute it returns, not an intrinsic property	DERIVED
9	The sphere has one aperture site; every polyhedron has more	COMPUTED
10	The sphere alone has generically zero-dimensional contact	DERIVED
10.1	Minimum boundary and minimum aperture are one property	DERIVED
11	Marks consume symmetry; the ledger saturates at three	THEOREM
11.1	An antipodal second mark costs nothing	DERIVED
12	Calculation requires distinctions transformation does not	DERIVED
12.1	Transformations exist that no mathematics can express	DERIVED
12.2	Mathematics is the trailing edge of transformation	DERIVED
13	Shape determines which mathematics is reachable	DERIVED
13.1	Lag sets select accessible structure; computed for two cases	COMPUTED
14	Matter is a stabilised transformation boundary; shape is its resolution	DERIVED

§	Claim	Grade
14.1	Form is not conserved; potential is	DERIVED
15	Every event exists as transformation and, at boundaries, as mathematics	DERIVED
15.1	The dual is asymmetric; transformation is prior	DERIVED
16	Transformations are selected from an already-complete set	DERIVED
16.1	Shape is memory: the past is the present admissible set	DERIVED
17	Continuation is a constraint on the query space, not on objects	DERIVED
18	The dynamical bridge is the next derivation owed	OPEN

20. Falsifiers

Each surviving claim is paired with an observation that would end it.

Claim	Would be falsified by
The sphere is the unique zero of δ	A non-spherical body with $\delta = 0$; the isoperimetric inequality's equality case forbids it.
Mathematics occurs only at boundaries	A geometric quantity of a homogeneous body definable without reference to its boundary.
The exterior carries no parameter	An intrinsic scalar of the unbounded component, defined without reference to the surface.
Content does not scale with extent	Two geometrically similar bodies of different size admitting different sets of operations.
The symmetry ledger saturates at three	A fourth mark on a sphere that opens a genuinely new coordinate rather than content within one.
Shape selects the mathematics	Two shapes supplying different coordinates but admitting identical operation sets.
Transformation precedes calculation	A calculation performable with no domain in which distinctions are held apart.
Transformations are selected	A transformation demonstrably not admissible for a shape before it occurred, becoming admissible by its occurrence.

The second entry is the load-bearing one. If any genuinely interior geometric quantity exists — one that cannot be rewritten as a boundary quantity — then interiors are not silent and the inversion fails at its root. The divergence theorem is the reason to expect none, since it converts volume integrals to boundary integrals whenever the integrand admits it, but a general proof that no interior quantity exists is not given here and is the paper's main open obligation.

21. Summary

Nothing is inside anything, in the sense that matters. Interiors are silent — not empty, but supplying no distinction and therefore no mathematics. Exteriors are nothing, carrying no parameter of their own. What remains is boundary, and boundary is where mathematics occurs, because contact is the only event that produces the relations mathematics is made of.

The sphere is the zero of this account. One scalar, measured inward. One aperture site. Zero isoperimetric deficit, uniquely. Contact with anything at a single point, minimally. It is not the simplest shape by convention — it is the shape that admits the least mathematics, and the isoperimetric inequality has been stating that for two centuries under a different reading.

Departure from the sphere is mathematical content. A faceted body has more boundary per unit of being, more sites where operations can enter, and correspondingly more that can be said about it. A body is not more substantial for being large. It is more substantial for being touched.

Transformation is prior to all of this and requires none of it. It occurs in interiors where nothing can be said about it, and it occurred before any relation existed to describe it. Mathematics is what grows at the boundaries transformation produces, always behind it, expanding only where a new contact has opened a place for it to expand into.

The word aperture carries its literal sense throughout. An aperture is not a hole information passes through. It is the boundary at which mathematical distinction becomes possible at all. An unmarked sphere has none. Contact opens the minimum one. Every further relation widens it, and a wider aperture admits more queries, which distinguish more transformations, which open further boundaries.

So the sequence from geometry through physics to biology and observation is not matter accumulating mathematics. It is the accumulation of places where mathematics can exist. [INTERPRETATION]

One process. Unconditional in itself, conditional in its sayability, and the condition is contact.